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Sub ject : DSA

Unit 3, S13, SLO2 - DEQUE - Advantages, Disadvantages and Applications

1. A deque allows ____ and ____ from both ends.
Answer: insertion and deletion

2. Deques are more ____ than standard queues for certain applications.
Answer: flexible / versatile

3. A deque can function as both a ____ and a ____.
Answer: stack and a queue

4. Deques provide efficient ____ complexity for insertion and deletion at both ends.
Answer: O(1) constant time

5. Deques are useful in implementing algorithms like ____ sliding window problems.
Answer: maximum / minimum

6. Deques require more complex ____ logic compared to simple queues.
Answer: pointer and index manipulation

7. Managing both ends in a deque can increase the chance of ____ errors.
Answer: implementation / boundary

8. Deques may require more ____ to track front and rear positions.
Answer: variables / memory

9. Linked list implementation of deques increases memory usage due to ____.
Answer: double pointers (prev and next)

10. Deques are used in ____ scheduling algorithms where elements are added or removed from both ends.
Answer: work-stealing / processor

11. In ____ problems, deques help maintain a sliding window of elements.
Answer: sliding window maximum

12. Deques can be used to implement both ____ and ____ data structures.
Answer: LIFO (stack) and FIFO (queue)